Distances problem

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Abstract

We show the following result, which reinforces the close connection between distances in a graph and trees. Given a connected graph G, $k \leq |V(G)|$, vertices $\{v_1, \ldots, v_k\} \subseteq V(G)$, and $F \subseteq E(G)$; assume that F is such that, when removed from $E(G)$, the distances between each pair of vertices $\{v_i, v_j\}$, $\{i, j\} \in \binom{[k]}{2}$ strictly increases: then $|F| \geq k - 1$.

1 Introduction

This is a problem which was communicated to us by [].

For each pair of vertices u, v in a graph G, let $d_G(u, v)$ denote the distance between u and v, that is, the minimal length (in terms of the number of edges) of a path between u and v, being set as ∞ if no path between u and v exists. For a set of edges F in $E(G)$, $G \setminus F$ denotes the graph on vertices $V(G)$ and edges $E(G) \setminus F$, and $V(F)$ denotes the vertices adjenct to and edge in F.[1]

Theorem 1. *For each connected graph G, each k with $k \in \{2, \ldots, |V(G)|\}$, each $\{v_1, \ldots, v_k\} \subseteq V(G)$ and each $F \subseteq E(G)$, if for each $\{i, j\} \in \binom{[k]}{2}$ we have that $d_G(v_i, v_j) < d_{G \setminus F}(v_i, v_j)$, then $|F| = s \geq k - 1$.*

Observe that there is no upper bound (at least, independent of $|V(G)|$), and that the lower bound is achieved, in among other examples, when G is a tree. The graph G can be assumed to be simple (without loops or multiple edges), for it is the worst case scenario as the loops do never become part of a minimal-length path, and, in a multiple edge we either remove the whole parallel class or we do not need to remove an edge of the parallel class at all, hence forming a smaller set in the underlying simple graph. Another way to see the set F in the statement of Theorem 1 is that, the edges of each minimal-length path between two vertices in $\{v_1, \ldots, v_k\}$ intersects F in a non-empty set.

Let us begin by showing that, used repeatedly, allows us to assume that some of the (actually all) the vertices from $\{v_1, \ldots, v_k\}$ are incident with edges in F.

Proposition 2. *Let $G, F, \{v_1, \ldots, v_k\}$ be with the properties given in Theorem 1. Then, if v_i is not in $V(F)$, the graph G' obtained from G by contracting all the edges incident to v_i, is such that the triple $G', F, \{v_1, \ldots, v_k\}$ also satisfies that, for each $\{j, j'\} \in \binom{[k]}{2}$, $d_{G'}(v_j, v_{j'}) < d_{G' \setminus F}(v_j, v_{j'})$.*

Proof of Proposition 2. Observe that no edge in F is contracted, as v_i is not in $V(F)$. Further, no neighbour of v_i belongs to $\{v_1, \ldots, v_k\} \setminus v_i$ for otherwise, the distance between v_i and some v_j would be one, and an edge in F would be between v_i and v_j contradicting the hypothesis that v_i is not adjacent to an edge in F. Now, any path from v_i to a v_j, $j \neq i$, uses an edge adjacent to v_i and gets as its first edge, and its length is reduced by 1; in particular, a minimal path from v_i to v_j in G is still a minimal path in G', but with the first edge removed (as it has been contracted) and this last edge does not belong to F). Thus, if F would

[1] That is, $V(F) = \{v \in V(G) \mid \exists u \in V(G), \exists e \in F, uv = e\}$.

destroy these paths in G, then they do it as well in G'. If a minimal path from v_j to v_r with $i \neq r$ in G is still a minimal path in G' from v_j to v_r, then it also gets destroyed when removing F in G'. Consider now a minimal path P in from v_j to v_r with $i \neq r$ in G'. If P does not go through v_i, then P was also a path in G and hence it contains an edge in F (as we have only contracted edges adjacent to v_i). On the other hand, if P goes through v_i, then both the subpaths from v_j to v_i, and from v_i to v_r are minimal-length paths with an end in v_i which, by what has been previously said, contain an edge in F. Thus the result follows. $\square$

Proof of Theorem 1. By induction on k. The result holds for $k = 2$, since at least one minimal path should be removed. Assume the result holds for a given $k \geq 2$, and let us show that then $s \geq (k+1)-1$ for each set of vertices of size $k+1$, each graph, and each set of edges F satisfying the incresed-distances-when-removed property.

Assume for a contradiction that: there exists a connected graph G, a set $|F| < k$, and a set of vertices $\{v_1, \ldots, v_{k+1}\}$ with the property that, after removing F, all the pairwise distances have increased. Using the procedure from Proposition 2 repeatedly,[2] we consider $G', F, \{v_1, \ldots, v_{k+1}\}$ be the triple satisfying $d_{G'}(v_i, v_j) < d_{G' \setminus F}(v_i, v_j)$, for each $\{i, j\} \in \binom{[k+1]}{2}$, and with the property that v_1 is one of the vertices of $V(F)$ (incident with an edge in F). Let e be the (an) edge incident with v_1.

We claim that the triple $G', F \setminus e, \{v_2, \ldots, v_{k+1}\}$ (removing e from the set of edges, and v_1 from the set of vertices) satisfies that $d_{G'}(v_i, v_j) < d_{G' \setminus [F \setminus e]}(v_i, v_j)$ for each $\{i, j\} \in \binom{\{2,\ldots,k+1\}}{2}$. We have two cases. If a minimal path P between v_i and v_j does not contain the edge e, then the edges in P that are in F belong to $F \setminus e$ and P is destoyed when $F \setminus e$ is removed. The second case occurs when P contains the edge e. Then $P = v_i P_1 v_1 e v' P_2 v_j$ or $P = v_i P_3 v' e v_1 P_4 v_j$ (depending on how the path P crosses the edge e), where P_1, P_2, P_3 and P_4 are paths (only considering their edges) with P_1 and P_4 non-empty paths (as $v_i, v_j \neq v_1$), and where v' is the other end of e besides v_1. Since P is a minimal path between v_i and v_j, then either P_1 is a minimal path between v_i and v_1 or, P_4 is a minimal path between v_1 and v_j. In both cases, since the edge e does not belong to P_1 nor to P_4, then we know that by removing the edges in $F \setminus e$ these minimal paths are also destroyed (in G'), and thus the path P contains an edge in $F \setminus e$ in it, and thus when removing $F \setminus e$, $P$is destroyed. Thus, the claim is shown.

Hence, by the induction hypothesis, we have that $|F \setminus e| \geq k-1$ which implies that $|F| \geq k$ as claimed. $\square$

[2]Note that some edges turn into multiple edges and loops when contracting other all the edges incident with v_i; if that would be the case, it may happen that some edges in F becomes useless, as they may turn into loops, for instance. Therefore, an even smaller (or equal in case these edges are not in F) set of edges can be assumed.